

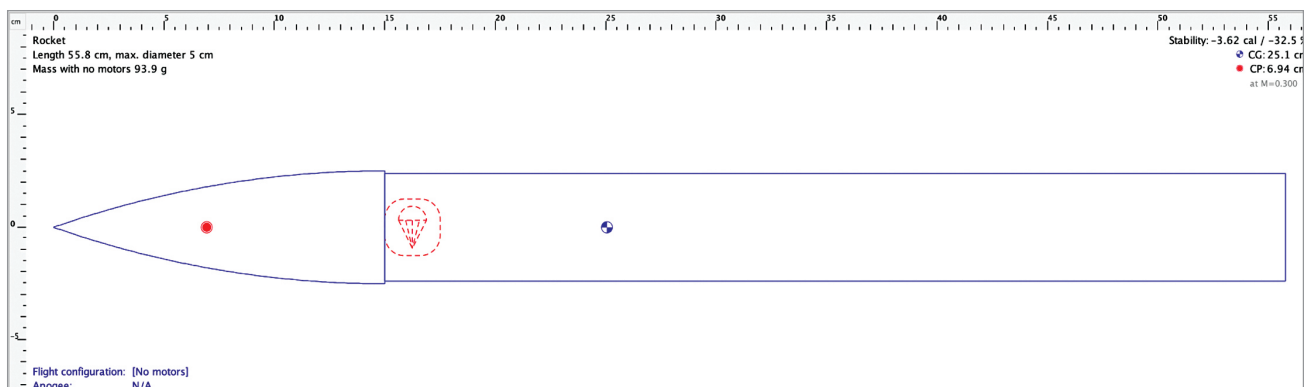


Fig. 3: Model rocket tube design

margin, and for most model rockets, it is recommended to keep it between 1 and 2 calibres for safe and stable flight.

Rocket design involves several important aspects, such as nose cone design, body tube selection, multi-stage rocket configuration, fin design, servo-based fin control, rocket engine selection, and flight simulation.

Nose design. The design process in OpenRocket begins with the topmost part of the rocket, known as the nose cone. Rocket nose cones are generally available in three common forms: Ogive, Elliptical, and Conical. In OpenRocket, a library of standard rocket components is already available, and these parts can be commercially sourced with ease. However, if a customised component is required, the specifications of each part can be modified according to the design requirements. The customised

parameters can then be exported as CAD files for manufacturing using CNC machining, 3D printing, or other additive manufacturing processes.

The nose cone shape significantly affects the aerodynamic performance and drag characteristics of the rocket. Fig. 2 shows the nose cone configuration and aerodynamic comparison in OpenRocket. It illustrates different nose cone profiles, along with their airflow interaction, drag characteristics, flow separation behaviour, and suitability for different rocket applications. For more details, refer to: <https://tuengr.com/V13/13A13J.pdf>.

The body tube. Once the nose cone design is understood, the next important part of rocket design is the body tube. The body tube serves as the main structural section of the rocket, housing all

major components, including the rocket motor, electronics, parachute system, payload, and wiring. Designing the body tube is not simply about creating a cylindrical structure. It directly affects the rocket's stability, aerodynamic drag, structural strength, and internal component arrangement.

One of the most important rules when designing the body tube is maintaining a proper length-to-diameter ratio. In most model rockets, the recommended ratio is:

$$\text{Length/diameter} = 10 \text{ to } 50$$

This means that if the rocket body diameter is 50mm, the total body length should ideally be between 500mm and 1000mm. If the rocket is too short, it may become unstable during flight, while an excessively long rocket can introduce structural bending, additional weight, and higher aerodynamic drag.

The diameter of the body tube mainly depends on the size of the motor, parachute system, and onboard electronics. For most beginner and intermediate model rockets, a diameter between 40mm and 60mm is commonly used.

Along with diameter, wall thickness is also important. A thicker body increases structural strength but also adds unnecessary mass, reducing flight efficiency.

Another critical aspect is the internal arrangement of components. The rocket motor is generally placed at the rear section, the electronics bay is positioned near

TABLE 2
ROCKET ELECTRONICS

Category	Component	Function	Quantity
Flight computer	IndusBoard Coin	IndusBoard Coin with stabilisation and flight computer firmware	1
Fin controller	Servo motor	Micro gear servo motor for model rocket fin control	3
Power	3.7V Li-Po battery	Li-Po battery for rocket circuitry power supply	1
Communication	Telemetry module	Module for remote rocket data transmission	1
Video	FPV camera	Camera for rocket launch and in-flight video transmission	1
Altitude sensing	BME280	Sensor for measuring air pressure, altitude, and temperature around the rocket	1
Motion sensing	MPU6050 gyro and accelerometer sensor	Sensor for orientation and acceleration data measurement	1

EFY note. Depending on simulation results and subsequent design modifications, additional components may be required in later stages. Fabrication materials such as screws, adhesives, and mounting hardware will also vary according to the selected rocket system design.